Coupled cusp constraints for three specified ideal right-angled four-facet types

Coxeter Polytopes Frontier Project

18 October 2026

Abstract

Consider a hypothetical entirely ideal right-angled hyperbolic five-polytope whose facets are restricted to the regular 24-cell, its facet double, and a particular sixteen-cell positive-carrier polytope. Cusp incidences and Lorentz orthogonality force at least 30 sixteen-cell facets and 66 doubled-24-cell facets. At a distinguished cusp of each sixteen-cell facet, a complete finite necessary-compatibility test forces eight sixteen-cell incident facets and excludes one of two possible box widths. One local labeling and an aggregate count assignment survive. Neither is a geometric realization; the three-type spectrum and the unrestricted ideal-five-dimensional problem remain open.

1 The specified facet family

Write C for the regular ideal right-angled 24-cell and D for its reflected double across an octahedral facet. Let T be the exact sixteen-cell polytope in the separate positive-carrier construction: starting with D, double across the successive facets with saved unit normals

$$F = (1, 0, 1, -1, 0), \quad G = (1, 0, 1, 0, -1), \quad H = (1, 0, 1, 1, 0).$$

The metric is $\mathrm{diag}(-1, 1, 1, 1, 1)$, and the construction's halfspace-union argument proves that these are actual finite-volume ideal right-angled polytopes. The full rational normal and vertex sets are the immutable supplementary inputs specified in the reproduction guide. No classification of all ideal right-angled four-polytopes is being assumed here.

For each ideal vertex $v = (1, x)$, the six incident unit normals form three opposite pairs. Each pair has sum λv, with $\lambda > 0$. In upper-half-space coordinates sending v to infinity, the horizontal normal has unit Euclidean length. Thus the horospherical width is proportional to λ, with one common scale for that cusp. Complete exact reconstruction gives

Facet	$(1,1,1)$ cusps	$(1,1,2)$ cusps	$(1,2,2)$ cusps	κ
C	24	0	0	0
D	36	6	0	-2
T	195	49	14	6

Here $\kappa = 3f_0 - 4f_3 + 24$. The three-facets of T have vertex-count histogram $6^{128}, 9^{48}, 14^{18}, 24^4$. Every cusp record retains the widths, opposite-pair assignments and the sizes of its six actual incident three-facets.

2 Global incidence and orthogonality

Theorem 1. *Suppose an entirely ideal finite-volume right-angled hyperbolic five-polytope has every facet isometric to C, D or T. If their numbers are x, y, z, respectively, then*

$$y = 3z - 24, \qquad 6 \mid z, \qquad z \geq 30, \qquad y \geq 66.$$

Proof. The separately proved ideal-five-dimensional carrier identity $\sum \kappa = 48$ gives $-2y + 6z = 48$. Every ambient cusp is a rectangular four-box. All its three-box facets have one of the three shapes in the table. Every pair of side lengths occurs in a triple, so all ratios are at most two; three distinct lengths would give an unlisted triple. The possible four-box shapes are therefore $1111, 1112, 1122, 1222$, up to scale and permutation. Write a, b, c, d for their numbers. Their eight facets have respective three-box type multiplicities

$$(8, 0, 0), \quad (2, 6, 0), \quad (0, 4, 4), \quad (2, 0, 6).$$

Counting actual cusp occurrences yields

$$\begin{aligned} 8a + 2b + 2d &= 24x + 36y + 195z, \\ 6b + 4c &= 6y + 49z, \\ 4c + 6d &= 14z. \end{aligned}$$

The difference of the last two gives $6(b - d) = 6y + 35z$, hence $6 \mid z$.

Each of the 22 three-facets of T with 14 or 24 vertices must meet another T: neither C nor D has such a facet. These 22 neighbors are distinct because the intersection of two convex facets is a single face. The ridge-adjacency graph on the z facets of type T therefore has at least $11z$ edges. It has no six-clique, since six mutually orthogonal unit spacelike vectors cannot occur in Lorentz space of positive index five.

An elementary weighted clique argument gives $e \leq 2z^2/5$ for a graph without a six-clique. Maximize $\sum_{ij \in E} t_i t_j$ for $t_i \geq 0$, $\sum t_i = 1$, choosing a maximizer with minimal support. If two supported vertices are nonadjacent, moving their combined weight to the one with larger weighted neighbor sum does not reduce the objective and reduces the support. Hence the support is a clique, of size at most five, and the objective is at most $(1 - 1/5)/2 = 2/5$. Uniform weights give the claim. Thus $11z \leq 2z^2/5$. Since $z > 0$ by $y = 3z - 24$, this implies $z \geq 55/2$. Divisibility by six now gives $z \geq 30$ and $y \geq 66$. □

3 A forced local eight-facet configuration

The complete cusp table of T has a unique vertex with three opposite ridge-size pairs

$$(24, 24), \quad (24, 24), \quad (14, 14),$$

whose widths are $(4, 4, 8)$ in the saved normalization. All six neighboring ambient facets at this vertex must be T. The ambient four-box has widths $(t, 4, 4, 8)$, where the original T is perpendicular to the t direction. A facet perpendicular to an old width-four direction has three-box widths $(t, 4, 8)$, so the allowed shape list forces $t \in \{4, 8\}$. The opposite of the original T has shape $(1, 1, 2)$, and must be D or T.

Theorem 2. *Under the same facet-spectrum restriction, the distinguished cusp has $t = 4$ and all eight incident ambient facets have type T.*

Proof. For each of the four cases $(t, \text{opposite type})$, take every actual cusp record of the required facet type. Allow all permutations and sign flips of the three coordinate axes preserving widths up to a common scale. On adjacent box facets, require equal vertex counts for their common three-face. This is a necessary overdomain: it even allows cusp symmetries which might not extend to the full four-polytope. The complete finite check gives

t	Opposite facet	Necessary matching
4	D	none
4	T	an explicit labeling
8	D	none
8	T	none

The exact domains and exhaustive negative checks are saved and replayed by two separately implemented searches. The producer uses direct backtracking; the standalone checker reconstructs the domains and uses arc consistency followed by complete branching. For $t = 8$, there is also a direct table obstruction: every $(1, 2, 2)$ cusp of T has nine-vertex three-facets on its four long-direction sides. A required neighbor has this cusp shape, but the side facing the original T is in the long t direction and has 24 vertices. The $t = 4, D$ exclusion uses the coupled adjacent-facet matching and is not inferred from the width list alone. □

4 Surviving counts and proof scope

The nonnegative assignment

$$(x, y, z) = (361, 66, 30), \qquad (a, b, c, d) = (2051, 241, 105, 0)$$

satisfies the displayed incidence equations and the ideal-five-dimensional Euler identities, with 457 facets and 2397 vertices. It also satisfies the elementary lower bound $1+198+258 = 457$ imposed by a T facet and its distinct ridge-neighbor and cusp-opposite facets. It is only an arithmetic witness. The surviving local labeling does not establish full three-face isometries, a consistent boundary complex, a Lorentz Gram realization or a chamber. Unlisted facet types are outside both theorems.

Generation, separate-process replay and the standalone checker passed. An independent full internal mathematical source/report/helper review checked the universal arguments and finite-domain scope, without re-executing them or reproving the inherited four-polytopes. No external human review or end-to-end formal-kernel proof is claimed. See `docs/ideal_ra5_cusp_compatibility_reproduce.md` for exact files, commands, immutable input hashes and the internal/external lemma map. The universal ideal-six-dimensional size and mixed-density consequences are presented separately in *Necessary size and face-density constraints for ideal right-angled six-polytopes*; they have no three-type facet-spectrum restriction.